

COVER PAGE

Pinnacle Logistics Solutions

Cybersecurity Gap Assessment Using the NIST Cybersecurity Framework (CSF) 2.0

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Company Overview

Company Overview

Organization Background

Pinnacle Logistics Solutions is a fictional third-party logistics (3PL) company headquartered in Houston, Texas. The organization provides warehousing, inventory management, freight coordination, and order fulfillment services for small and medium-sized businesses across the southern United States.

The company employs approximately 120 employees, including warehouse personnel, office staff, remote sales representatives, and executive leadership. Daily business operations depend on secure access to customer information, inventory systems, financial records, and cloud-based collaboration tools.

Because the company manages sensitive customer information and relies heavily on technology to support logistics operations, maintaining a strong cybersecurity posture is essential to protecting business continuity, customer trust, and regulatory compliance.

Business Assets

The following assets are considered critical to the organization's operations:

- Customer shipping and logistics records
- Employee personnel information
- Financial and accounting data
- Warehouse Management System (WMS)
- Microsoft 365 environment
- Windows workstations and laptops
- Windows Server infrastructure
- Company VPN
- Shared file servers
- Wireless network infrastructure

Technology Environment

Pinnacle Logistics Solutions currently operates the following technology environment:

- Windows 11 endpoints
- Windows Server
- Microsoft 365
- Microsoft Outlook
- Microsoft Teams
- OneDrive
- Microsoft Defender for Business
- Company VPN
- Enterprise Wi-Fi
- Warehouse Management System (WMS)

Business Objectives

The organization's primary cybersecurity objectives include:

- Protect sensitive customer and employee information.
- Maintain continuous warehouse and logistics operations.
- Reduce cybersecurity risk through effective security controls.
- Detect and respond to security incidents in a timely manner.
- Improve overall cybersecurity maturity using the NIST Cybersecurity Framework (CSF) 2.0.

Assessment Scope

Assessment Scope

The purpose of this cybersecurity gap assessment is to evaluate Pinnacle Logistics Solutions' current cybersecurity posture using the National Institute of Standards and Technology (NIST) Cybersecurity Framework (CSF) 2.0.

This assessment examines the organization's cybersecurity governance, asset management practices, identity and access management controls, data protection measures, security monitoring capabilities, incident response preparedness, and recovery planning. The evaluation focuses on identifying gaps between the organization's existing cybersecurity controls and the security outcomes recommended by the NIST Cybersecurity Framework.

The assessment is based on a review of the organization's business operations, technology environment, and assumed security practices typical of a mid-sized third-party logistics provider. The findings are intended to identify organizational risks, prioritize remediation efforts, and provide practical recommendations that strengthen cybersecurity resilience while supporting business objectives.

This assessment is intended as a simulated portfolio project and does not represent an evaluation of an actual organization.

Assessment Objectives

Assessment Objectives

The primary objective of this cybersecurity gap assessment is to evaluate Pinnacle Logistics Solutions' current cybersecurity posture and determine its alignment with the National Institute of Standards and Technology (NIST) Cybersecurity Framework (CSF) 2.0. The assessment aims to identify existing cybersecurity strengths, evaluate control deficiencies, assess organizational risk, and provide practical recommendations that improve the organization's overall security maturity.

Specifically, this assessment seeks to

- Evaluate the effectiveness of the organization's current administrative, technical, and operational security controls.
- Assess the organization's cybersecurity capabilities across the six NIST CSF 2.0 functions: Govern, Identify, Protect, Detect, Respond, and Recover.
- Identify security gaps that could negatively impact the confidentiality, integrity, and availability of organizational information and critical business systems.
- Analyze cybersecurity risks based on their likelihood of occurrence and potential business impact.
- Prioritize remediation efforts using a structured risk-based approach.
- Develop actionable recommendations that strengthen cybersecurity governance, improve operational resilience, and support long-term business objectives.
- Demonstrate the practical application of cybersecurity governance, risk management, and compliance (GRC) principles through a structured organizational assessment.

Assessment Methodology

Assessment Methodology

This cybersecurity assessment was conducted using the NIST Cybersecurity Framework (CSF) 2.0 as the primary evaluation framework. The assessment focused on six core cybersecurity functions: Govern, Identify, Protect, Detect, Respond, and Recover.

Information used during the assessment was based on a review of the organization's documented technology environment, existing security controls, business operations, and assumed organizational practices typical of a mid-sized third-party logistics company.

Each function was evaluated to determine the effectiveness of current security controls, identify control gaps, assess organizational risk, and recommend practical improvements. Risks were prioritized based on their likelihood of occurrence and potential impact to business operations.

Current Security Environment

Current Security Environment

Overview

As part of this assessment, Pinnacle Logistics Solutions' existing cybersecurity environment was evaluated to determine the effectiveness of current security controls and identify opportunities for improvement. The organization has implemented several foundational cybersecurity measures that provide a baseline level of protection for its information systems and business operations. However, the assessment identified several areas where additional controls and governance practices are needed to better align with the NIST Cybersecurity Framework (CSF) 2.0.

Existing Security Controls

The organization currently maintains the following cybersecurity controls:

- Microsoft Defender for Business is deployed across all company-managed Windows endpoints to provide endpoint protection against malware and other common threats.
- Microsoft 365 is used as the organization's primary cloud platform for email, collaboration, and file storage through Outlook, Teams, and OneDrive.
- Employees access internal business resources remotely through a company-managed Virtual Private Network (VPN).
- Windows operating systems receive routine security updates and software patches through centralized update management.
- Password policies enforce minimum complexity requirements for employee accounts.
- Network firewalls are configured to help restrict unauthorized network traffic.
- Microsoft 365 data is backed up regularly to support business continuity and data recovery.
- Employees participate in annual cybersecurity awareness training covering phishing, password security, and general cybersecurity best practices.

Identified Security Gaps

Although the organization has implemented several foundational controls, the assessment identified multiple opportunities to improve its overall cybersecurity maturity.

The most significant findings include:

- Multi-factor authentication (MFA) has not been fully implemented across all business applications and remote access services.
- Hardware, software, and cloud asset inventories are incomplete and are not reviewed on a recurring basis.
- Security logs are generated but are not centrally collected, correlated, or monitored through a Security Information and Event Management (SIEM) platform.
- A formal Incident Response Plan has not been developed, documented, or tested.
- Vulnerability assessments and penetration testing are not conducted on a scheduled basis.
- Third-party vendors are not evaluated using a formal cybersecurity risk assessment process before being granted access to organizational systems or data.

- Security awareness training is conducted annually but lacks continuous reinforcement through periodic awareness campaigns or phishing simulations.
- The organization does not currently measure cybersecurity maturity against an established framework or maintain cybersecurity performance metrics for executive leadership.

Overall Assessment

Based on the current security environment, Pinnacle Logistics Solutions demonstrates a developing cybersecurity program with several effective foundational controls already in place. However, improvements to governance, identity and access management, continuous monitoring, incident response, and risk management are necessary to strengthen the organization's security posture and improve alignment with the NIST Cybersecurity Framework (CSF) 2.0.

NIST CSF 2

NIST CSF 2.0 Gap Assessment

NIST CSF Function	Current State	Gap Identified	Risk Level
Govern	Partial	Security governance policies are informal and not consistently reviewed.	Medium
Identify	Partial	Asset inventory is incomplete and not regularly updated.	High
Protect	Partial	MFA is not fully implemented, and security awareness training occurs only annually.	High
Detect	Limited	Security logs are not centrally monitored, and alerting capabilities are minimal.	High
Respond	Limited	No formal incident response plan or documented response procedures exist.	High
Recover	Partial	Backups exist, but disaster recovery testing is not performed regularly.	Medium

Govern

The Govern function focuses on establishing cybersecurity governance, defining organizational responsibilities, and ensuring cybersecurity activities support business objectives.

Current State

Pinnacle Logistics Solutions has basic cybersecurity policies and management oversight; however, governance activities are informal, and cybersecurity responsibilities are not consistently documented across the organization.

Gap Identified

The organization lacks a structured cybersecurity governance program, formal policy review schedule, and clearly defined governance metrics.

Risk Level: Medium

Identify

The Identify function focuses on understanding organizational assets, business processes, and cybersecurity risks.

Current State

The organization maintains basic documentation of hardware and software assets but does not maintain a complete or regularly updated inventory.

Gap Identified

Incomplete asset inventories increase the likelihood that vulnerable or unauthorized devices remain unmanaged.

Risk Level: High

Protect

The Protect function focuses on safeguarding systems, users, and organizational data.

Current State

Password policies, Microsoft Defender, VPN access, and employee security awareness training are currently implemented.

Gap Identified

Multi-factor authentication has not been fully deployed across business systems, and security awareness training occurs only once each year.

Risk Level: High

Detect

The Detect function focuses on identifying cybersecurity events before they become major incidents.

Current State

Endpoint antivirus generates alerts, but centralized security monitoring and log analysis capabilities are limited.

Gap Identified

The organization lacks centralized log monitoring and proactive detection capabilities.

Risk Level: High

Respond

The Respond function focuses on containing and managing cybersecurity incidents.

Current State

IT personnel respond to incidents as they occur; however, there is no documented incident response plan.

Gap Identified

Without standardized procedures, incident response activities may be inconsistent and increase recovery time.

Risk Level: High

Recover

The Recover function focuses on restoring business operations following a cybersecurity incident.

Current State

Microsoft 365 backups are performed regularly; however, disaster recovery procedures are not routinely tested.

Gap Identified

Recovery plans may not perform as expected during an actual ransomware attack or system outage because they have not been validated through testing.

Risk Level: Medium

Risk Register

Risk Register

The cybersecurity assessment identified ten primary risks that may negatively affect Pinnacle Logistics Solutions if left unaddressed. Each risk was evaluated based on its likelihood of occurring and the potential business impact. Risks involving identity and access management, asset management, security monitoring, and incident response received the highest priority due to their potential to significantly disrupt business operations and expose sensitive information.

The identified risks were documented in a formal Risk Register to support risk tracking, remediation planning, and executive decision-making. Recommendations were prioritized according to their potential impact on improving the organization's overall cybersecurity maturity.

Risk Matrix

Risk Matrix

Risk ID	Risk	Likelihood	Impact	Overall Rating
R-001	MFA not fully implemented	High	High	● High
R-002	Asset inventory incomplete	High	High	● High
R-003	No centralized log monitoring	High	High	● High
R-004	No Incident Response Plan	Medium	High	● High
R-005	Disaster recovery not tested	Medium	High	● Medium
R-006	Annual security training only	Medium	Medium	● Medium
R-007	No vendor risk assessments	Medium	High	● Medium
R-008	No regular vulnerability scans	High	High	● High
R-009	No periodic access reviews	Medium	Medium	● Medium
R-010	No cybersecurity metrics	Low	Medium	● Low

The Risk Matrix provides a visual representation of the organization's highest-priority cybersecurity risks. Risks were evaluated using two criteria: the likelihood of occurrence and the potential impact on business operations. Risks rated as "high" require immediate remediation because they present the greatest threat to the confidentiality, integrity, and availability of organizational information. Medium risks should be addressed through planned improvements, while low risks should continue to be monitored as the organization's cybersecurity program matures.

Recommendations

RECOMMENDATIONS

Based on the assessment findings, the following recommendations are prioritized according to their potential impact on reducing organizational risk and improving cybersecurity maturity.

High Priority Recommendations

1. Implement Multi-Factor Authentication (MFA)

Require MFA for Microsoft 365, VPN access, and all critical business applications to reduce the risk of unauthorized account access and credential compromise.

2. Develop a Formal Incident Response Plan

Create, document, and test an Incident Response Plan that clearly defines roles, responsibilities, communication procedures, and recovery activities during cybersecurity incidents.

3. Implement Centralized Security Monitoring

Deploy a centralized Security Information and Event Management (SIEM) solution, such as Microsoft Sentinel or Splunk, to improve visibility into security events and enable faster detection of suspicious activity.

4. Establish a Complete Asset Inventory

Maintain a centralized inventory of all hardware, software, and cloud assets to improve vulnerability management, patching, and risk identification.

Medium Priority Recommendations

5. Conduct Quarterly Vulnerability Assessments

Perform routine vulnerability scans and periodic penetration testing to identify security weaknesses before they can be exploited.

6. Improve Security Awareness Training

Replace annual training with quarterly security awareness sessions and phishing simulations to strengthen employee security awareness.

7. Establish a Third-Party Risk Management Process

Evaluate vendors before onboarding and periodically reassess third-party security practices to reduce supply chain risk.

Low-Priority Recommendations

8. Develop Cybersecurity Metrics and Reporting

Create executive dashboards that measure cybersecurity performance using key metrics such as patch compliance, phishing results, incident response times, and vulnerability remediation progress.

Implementing these recommendations will improve the organization's alignment with the NIST Cybersecurity Framework (CSF) 2.0, strengthen its cybersecurity posture, and reduce overall organizational risk.

Executive Summary

Executive Summary

Pinnacle Logistics Solutions requested a cybersecurity gap assessment to evaluate its current security posture using the National Institute of Standards and Technology (NIST) Cybersecurity Framework (CSF) 2.0. The objective of this assessment was to identify weaknesses in existing cybersecurity controls, evaluate organizational risks, and provide recommendations that improve the company's overall cybersecurity maturity.

Overall, the organization demonstrates a solid foundation of cybersecurity controls, including endpoint protection, password policies, Microsoft 365 security features, VPN access, cloud backups, and annual security awareness training. These controls provide a reasonable baseline for protecting business operations.

However, the assessment identified several high-priority gaps that increase organizational risk. The most significant findings include the absence of organization-wide multi-factor authentication (MFA), limited centralized security monitoring, incomplete asset inventory management, and the lack of a formal Incident Response Plan. Additional opportunities for improvement include implementing regular vulnerability assessments, strengthening third-party risk management, and increasing the frequency of security awareness training.

Based on the assessment findings, this report recommends prioritizing identity and access management improvements, enhancing security monitoring capabilities, strengthening governance processes, and developing formal incident response procedures. Implementing these recommendations will improve the organization's alignment with the NIST Cybersecurity Framework (CSF) 2.0 while reducing overall cybersecurity risk and improving operational resilience.

Conclusion

This cybersecurity gap assessment evaluated Pinnacle Logistics Solutions against the NIST Cybersecurity Framework (CSF) 2.0 to identify opportunities for improving its cybersecurity program. While the organization has implemented several foundational security controls, additional improvements are necessary to reduce cyber risk and strengthen long-term resilience.

The assessment identified several high-priority areas requiring immediate attention, including multi-factor authentication, centralized security monitoring, incident response planning, and comprehensive asset management. Addressing these recommendations will significantly improve the organization's ability to protect critical information, detect malicious activity, respond effectively to cybersecurity incidents, and recover from disruptive events.

Cybersecurity is a continuous process rather than a one-time activity. Regular assessments, ongoing employee training, continuous monitoring, and periodic reviews of organizational security controls will help Pinnacle Logistics Solutions maintain a mature cybersecurity program aligned with industry best practices and evolving threats.